SARCLIP: The First Vision–Language Foundation Model for SAR Image

Pengfei Wang, Zhuhao Lu, *Student Member, IEEE*, Yajun Li, Baogang Ding, and Ding Zhang

***Abstract*—Foundation models have achieved remarkable breakthroughs across various domains, with the widely use of masked image modeling (MIM) and self-supervised learning (SSL). However, these models lack language comprehension, which limits their ability to generalize in downstream tasks and zero-shot applications. The contrastive language–image pretraining (CLIP) model overcomes these challenges by integrating textual information and performs exceptionally in remote sensing. However, no such multimodal model exists for the synthetic aperture radar (SAR) domain. To bridge this gap, we propose SARCLIP, a vision–language model specially designed for SAR, which leverages textual supervision to enhance visual representation and achieves fine-grained vision–language alignment through modality alignment. First, to overcome the lack of SAR image-to-text datasets, we propose the detection-to-caption (D2C) algorithm, which transforms the heterogeneous format of object detection into diverse captions, ultimately creating the vision–language dataset, MMSAR. In addition, to address the distortion and information loss caused by downsampling in high-resolution SAR images, we propose AnyReader, a dynamic module capable of processing images of any size through a low-resolution encoder. We validate SARCLIP's generalization and performance on five downstream tasks: image-to-text retrieval, zero-shot classification, few-shot classification, linear probing, and object counting. Especially, we introduce a new counting benchmark, SARCount. Our results demonstrate that SARCLIP consistently outperforms the CLIP baseline across various model sizes. Impressively, SARCLIP achieved a 7.07% higher average recall on the retrieval task compared to the largest CLIP model. In zero-shot classification, SARCLIP surpasses CLIP by 7.8% in average accuracy across three downstream datasets.**

***Index Terms*—Multimodal, SARCLIP, vision–language foundation model.**

I. Introduction

GENERAL-PURPOSE foundational models are becoming increasingly important in the field of artificial intelligence [1]. Unlike specialized small models that are limited to specific domains and require retraining with new datasets whenever the scenario changes, significantly increasing labor and time costs, foundational models offer a "once-and-for-all" solution. These models demonstrate exceptional performance and generalization across a wide range and diverse downstream tasks. Relying solely on large-scale pretraining and highly parameterized adaptive learning capabilities, they can become experts in a specific domain.

In recent years, many general-purpose models have emerged. In computer vision, SAM [2] enables flexible segmentation of any object or region in images, MAE [3] learns powerful visual representations through self-supervised learning (SSL) for tasks like object detection, image segmentation, and generation, while SimCLR [4] extracts valuable image features from large unlabeled datasets without the need for labels. In natural language processing, models, such as BERT [5], GPT [6], and LLaMA [7], leverage large-scale pretraining to learn complex language patterns and structures, excelling in text generation and understanding with broad applications across domains. In multimodality, GPT-4 [8] and QwenVL [9] combine language and visual understanding, enhancing machine perception of the real world. In addition, DALL-E [10] enables text-to-image generation, while Sora [11] marks a major breakthrough in text-to-video generation.

Remote sensing has also benefited from these advances, mainly through SSL methods such as masked image modeling (MIM) [3]. Studies like DINO [12], SatMAE [13], ViTAE [14], and GFM [15] applied MIM to large-scale remote sensing imagery and Vision Transformers, showing promising results. However, limitations of MIM-based approaches have become apparent. Prior work shows that such models mainly learn occlusion-invariant [16] and low-level features [17], [18], [19], which are relatively effective for dense prediction tasks like detection and segmentation but insufficient for high-level semantic tasks such as zero-shot or few-shot classification. Moreover, MIM tends to capture fine-grained textures rather than long-range dependencies, reducing robustness and generalization [20].

These shortcomings indicate that simply extending MIM is insufficient for building strong foundation models in remote sensing. In practice, existing models also require large amounts of manual labels and fine-tuning for different tasks, which hinders scalability. This motivates the shift toward vision–language pretraining. The contrastive language–image pretraining (CLIP) [21], for instance, demonstrates how language supervision can inject high-level semantics into visual representations and achieve strong zero-shot generalization. Multimodality plays a key role in foundational models for remote sensing [22]. While variants such as RemoteCLIP [23] and GeoRSCLIP [24] have validated this idea in optical

Received 23 January 2025; revised 28 October 2025; accepted 2 November 2025. Date of publication 7 November 2025; date of current version 19 November 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62371192. *(Corresponding author: Pengfei Wang.)*

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Digital Object Identifier 10.1109/TGRS.2025.3630131

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remote sensing, no such vision–language foundation model exists for synthetic aperture radar (SAR) imagery, where data scarcity and high-resolution distortion present unique and urgent challenges.

1) SAR data are scarce, with only object detection datasets available. The bounding box annotation format is hard for text encoders based on natural language to process, and large amounts of image-to-text paired data are required.
2) SAR images often have high resolution, and downsampling is required to match the size of the image encoder. However, this process causes image distortion and information loss, making effective image representation encoding challenging.

To address the aforementioned issues, this article proposes a visual–language foundational model: SARCLIP. First, we curate a comprehensive collection of open-source SAR datasets, with SARDet-100K [25] comprising ten major data subsets. We introduce the detection-to-caption (D2C) algorithm to convert heterogeneous formats of object detection into unified textual representations, which primarily include information such as category, quantity, and relative position. We also focus on diversifying the text expressions to improve generalization. This process results in the MMSAR, the first paired image-to-text dataset in the SAR domain. Second, to address the challenge of effectively extracting features from high-resolution images, we propose the AnyReader architecture. By employing adaptive cropping, abstractor and compression, and a 2-D position embedding module, AnyReader can effectively extract features from high-resolution images using a low-resolution encoder, achieving fine-grained image-to-text alignment. We train the model to optimize the InfoNCE loss, aiming to maximize the mutual information between vision and language. After pretraining, the SARCLIP model is evaluated for its performance and generalization across five major downstream tasks: zero-shot classification, image-to-text retrieval, few-shot classification, and linear probing. Especially, we introduce a novel benchmark, SARCount, to validate the capability of target counting. SARCLIP is trained with various backbone architectures and model sizes, ranging from the 38M parameter ResNet50 to the 340M parameter ViT-L-16. Comprehensive evaluation on downstream datasets demonstrates that SARCLIP significantly outperforms the baseline of CLIP.

The contributions of this article can be listed as follows.

1) *Large-Scale Image-to-Text Dataset for SAR:* This article proposes a D2C algorithm to transform the heterogeneous object detection format of SARDet-100K into diverse textual descriptions, resulting in MMSAR, the first image-to-text paired dataset in the SAR domain.
2) *Innovative Visual–Language Foundational Model for SAR:* We propose an innovative visual–language foundational model: SARCLIP. Also, we propose the AnyReader architecture to dynamically handle images of any resolution. With large-scale data pretrained, the model achieves fine-grained visual–language alignment and leverages rich semantic supervision to learn robust visual representations.
3) *Extensive Downstream Task Applications for SAR:* The effectiveness of SARCLIP is validated across five downstream tasks: image-to-text retrieval, linear probing, zero-/few-shot classification, and counting. Notably, we introduce a new benchmark, SARCount, to evaluate the model's target counting capability.

II. Related Work

A. Vision–Language Models in Remote Sensing

The alignment and fusion of image-to-text multimodal information have been a longstanding research hotspot and challenge in the field of artificial intelligence [26], [27]. The challenge is also of great significance to the field of remote sensing, as supervised learning through natural language can better extract image features and provide effective interpretation of complex satellite images.

Multimodal Retrieval Models in Remote Sensing: The early work in remote sensing retrieval was led by Xu [28] and Lu et al. [29], who utilized CNNs for image encoding and LSTMs for text captions encoding. To avoid focusing solely on global information while neglecting local object features, Yuan et al. [30] proposed a multilevel information dynamic fusion module to effectively combine features from different levels. To fully leverage the cues from mismatched text fragments, a negative perceptual attention framework was proposed by Zhang et al. [31], which leverages the positive effects of matching fragments and the negative effects of mismatched fragments to jointly infer the similarity between images and text. A multilanguage encoder was proposed by Rahhal et al. [32], which enables satellite images to adjust to diverse languages. Subsequently, many studies have been proposed to align language and image features, leveraging the strong supervision of natural language to obtain knowledge representations for image-to-text retrieval models, such as HyperMatch [33], HVSA [34], CMFM-Net [35], KCR [36], and others. However, these models have only been validated for the specific task of image-to-text retrieval, and the performance on other downstream tasks remains unknown.

CLIP Model: CLIP is a multimodal model proposed by OpenAI [21]. It uses contrastive learning on 400M text-to-image pairs to minimize the distance between matching images and text, thereby effectively aligning vision and language representations. This approach learns highly transferable representations that generalize well to various downstream tasks and demonstrates strong zero-shot capabilities. The training image-to-text pairs mainly come from the internet and are not manually annotated, closely aligning with the self-supervised training paradigm of large-scale models. Current optimizations for the CLIP model mainly include model scaling law [37], data scaling [38], self-supervision [39], few-shot fine-tuning [39], and pretraining efficiency [40].

CLIP-Based Model: Due to its strong ability to represent multimodal features and zero-shot generalization, the CLIP model is widely used as a base model in research across various fields. For instance, MedCLIP [41], PubMedCLIP

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TABLE I

SOURCE DATASET DETAILS FOR SARDET-100K. S-1 (SENTINEL-1) AND GF-3 (GAOFEN-3). CATEGORIES: H: HARBOR; A: AIRCRAFT; S: SHIP; C: CAR; T: TANK; AND B: BRIDGE

Datasets	Res.(m)	Target	Band	Polarization	Satellites
Ship Dataset	3~25m	S	C	HH,VV, VH,HV	S-1,GF-3
SSDD	1~15m	S	C/X	HH	S-1,RandarSat-2,TerraSAR-X
MSAR	≤1m	A,B,T, S	C	HH,HV, VH,VV	HISEA-1
SADD	0.5~3m	A	X	HH	TerraSAR-X
HRSID	0.5~3m	S	C/X	HH,VV, VH,HV	S-1B,TerraSAR-X,TanDEMX
SAR -AIRcraft	1m	A	C	Uni-polar	GF-3
AIR_ SARShip	1,3m	S	C	VV	GF-3
OGSOD	3m	B,T,H	C	VV/VH	GF-3
SIVED	0.1m,0.3m	C	Ka,Ku,X	VV/HH	Airborne SAR synthetic slice

TABLE II

SARDET-100K DATASET: STATISTICS OF IMAGE AND INSTANCE

Datasets	Images				Instances				Ins/Img
	Train	Val	Test	All	Train	Val	Test	all	
ShipDataset[47]	31,784	3,973	3,972	39,729	40.761	5,080	5,044	50,885	1.28
SSDD[48]	928	116	116	1,160	2,041	252	294	2,587	2.23
MSAR[49]	27,159	1,479	1,520	30,158	58,988	3,091	3,123	65,202	2.16
SADD[50]	795	44	44	883	6,891	448	496	7,835	8.87
HRSID[51]	3,642	981	981	5,604	11,047	2,975	2,947	16,969	3.03
SAR-AIRcraft[52]	13,976	1,923	2,989	18,888	27,848	4,631	5,996	38,475	2.04
AIR_SARShip 1*[53]	438	23	40	501	816	33	209	1,058	2.11
AIR_SARShip 2*[53]	270	15	15	300	1,819	127	94	2,040	6.80
OGSOD[54]	14,664	1,834	1,833	18,331	38,975	4,844	4,770	48,589	2.65
SIVED[55]	837	104	103	1,044	9,561	1,222	1,230	12,013	11.51
SARDet-100k	94,493	10,492	11,613	116,598	198,747	22,703	24,023	245,653	2.11

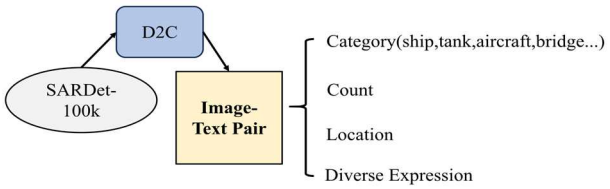


Fig. 1. Flowchart of D2C algorithm.

[42], and BiomedCLIP [43] have demonstrated outstanding application capabilities in the medical field. In the domain of remote sensing, RemoteCLIP [23], RS-CLIP [44], and GeoRSCLIP [24] have also achieved consistently positive outcomes. However, in the SAR image domain, no image-to-text retrieval models have been proposed, and there is no formal image-to-text paired data. The performance on downstream tasks remains unknown. Therefore, this article introduces the SARCLIP model to evaluate the effectiveness of using text information to supervise and guide SAR image processing.

Despite these advances, no large-scale paired dataset or CLIP-based model exists for SAR imagery. Current SAR datasets are mainly designed for object detection, and their bounding box annotations are unsuitable for direct multimodal training. Moreover, high-resolution SAR data poses unique challenges due to distortion and information loss from downsampling. These gaps highlight the necessity of a dedicated SAR vision–language foundation model, motivating our proposed SARCLIP framework.

B. SARDet-100K

The SAR field faces challenges due to the limited availability of public datasets and the lack of diversity in individual dataset samples. FUSAR-Ship 1.0 [45] and SRSDD-SAR [46] are two widely used SAR datasets. The former, created by Chinese Academy of Space Technology using GF-3 satellite imagery, contains 15 main ship categories, including ship images matched with AIS information, suitable for ship detection and identification under complex maritime conditions. However, the dataset is small, focusing mainly on ship targets, with a lack of category diversity, which limits its applicability in more complex detection tasks. Similarly, the SRSDD-SAR dataset, also created by the same institution, contains ship samples from 30 SAR images, with rotated bounding boxes for target labeling. Although it is suitable for ship target detection, it is limited by a small number of categories and a smaller data volume, making it less suited for large-scale multimodal tasks and model training. In contrast, SARDet-100K [25] addresses these limitations and is the first COCO-level multiclass, large-scale SAR object detection dataset. The data come from different countries and organizations, with no category conflicts. Table I describes the detailed information of the subdatasets, including imaging resolution, radar bands, four polarization modes, and satellite sources. Leveraging these high-quality SAR image sources, this article proposes a D2C algorithm to convert heterogeneous object detection formats into diverse textual captions. As shown in Table II, the dataset contains 245 653 instance targets and 116 598 images, with a total of six different categories: ports, airplanes, cars, ships, tanks, and bridges, thereby providing a comprehensive foundation for multimodal pretraining and evaluation across diverse operational scenarios.

III. PROPOSED METHODS

A. D2C Algorithm

In SAR imagery, datasets with image-to-text pairs are unavailable. However, object detection datasets like SARDet-100K provide bounding box and category annotations for detector training. CLIP cannot directly interpret these annotations as it is pretrained on natural language. Therefore, we propose a D2C algorithm to convert heterogeneous object detection boxes and categories into unified captions, aiming to effectively bridge this critical gap in multimodal learning. As shown in Fig. 1, each text describes the category, relative position, and quantity of the targets. To ensure data diversity,

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we paid special attention to varied expressions during text construction, such as using numerical descriptions in both Chinese and English, active and passive voice, and synonym replacements. Algorithm 1 provides a detailed explanation of the pseudocode for the D2C algorithm. Each image is categorized as either a single-class or a multiclass image. For single-class images, three descriptive sentences are generated for each image. Specifically, the first two captions are determined by the object locations (the centers of bounding boxes). The first caption describes the features of targets in the central region, while the second focuses on those in the peripheral region. The regional description provides additional spatial distribution context and content-rich details of the objects. The third caption captures the overall features of the targets in the entire image, complementing and enhancing the coherence of the first two captions.

Algorithm 1 D2C: Caption Generation From Detected Objects

Input: bounding boxes all box list; category–count map object.items()
Output: list of generated captions all captions

```
all captions ← []
Step 1: Single-category image
obj ← FindObjectInCenter(all box list)
append(GeneCaption(obj, "in the center of the image."))
obj ← FindObjectInEdge(all box list)
append(GeneCaption(obj, "at the edge of the image."))
append(GeneCaption(all box list, "in the picture."))
Step 2: Multi-category image
for i = 1 to 3 do
    rand obj ← RandomChoice(list(object.items()))
    append(GeneCaption(rand obj, "in the image."))
end for
procedure GeneCaption(obj, suffix)
    caption ← "There are "
    for all (cls, num) ∈ obj.items() do
        if num × Random() >0.8 then
            w ← RandomChoice(["a lot of", "many", "a quantity of"])
            caption ← caption + w + cls
        else
    caption ← caption + num + cls
        end if
    end for
    return caption + suffix
end procedure
```

For multiclass images, we generate 3–4 sentences of text descriptions. First, one class is randomly selected from multiple categories, and its features are described individually, with this process repeated 2–3 times. Then, we provide an overall description of all categories in the image, such as "there are three ships and one harbor in the SAR image." For scenarios with more than ten targets, we set a random seed. If the seed is greater than 0.8, quantifiers like "lots of" or "many" are used instead of specific numbers to enhance variability and readability.

B. Image-to-Text Dataset of MMSAR

MMSAR: Based on the D2C algorithm, we converted heterogeneous object detection annotations into diverse textual descriptions, generating a total of 304K image-to-text pairs across 105 685 images. As shown in Fig. 2, all captions are of high quality and diversity. Our training set primarily includes eight detection subsets from SARDet-100K: SAR-Ship-Dataset [47], Air-SARShip1 [53], Air-SARShip2 [53], HRSID [51], OGSOD [54], SADD [50], SAR-AIRcraft [52], and MSRA [49]. For downstream tasks, we selected SSDD [48], SIVED [55], and the supplementary SAR-Airport [56]. To validate the generality and adaptability of SARCLIP, we conducted evaluations on six downstream tasks, especially including our newly proposed SARCount benchmark for object counting. For further details, please refer to the experimental section.

Image Deduplication: The MMSAR dataset and downstream benchmarks consist of subsets from different sources. To prevent data leakage, it is necessary to remove similar samples between training and testing subsets, except within the same subset. We use the pHash [57] algorithm for image encoding and similarity calculation. Images are downsampled to 64×64 and converted to grayscale to reduce complexity and standardize inputs. The discrete cosine transform (DCT) converts images to the frequency domain, retaining only the top-left 8×8 low-frequency components, which preserve stable overall structure and brightness features. The average value of the DCT results is used to generate a binary hash. Values above the average are set to 1, and others to 0, forming a binary hash code representing the image. Similarity between images is measured by the Hamming distance of their pHash values—the smaller the distance, the more similar the images.

Data Analysis: Fig. 3 shows the text length distribution in MMSAR, with descriptions mostly ranging from 9 to 11 fields per image. Fig. 4 visualizes the category distribution across 222 120 instances in seven major categories: ships, bridges, aircraft, tanks, cars, harbors, and airports. Ships and aircraft dominate, while airports have the fewest instances.

C. SARCLIP

Trained on high-quality SAR image-to-text pairs, SARCLIP exhibits strong cross-modal representation capabilities. As shown in Fig. 5, it consists of three components: an adaptive dynamic resolution image encoder, a text encoder, and an image-to-text contrastive learning matrix. Given a batch of N image-to-text pairs, SARCLIP jointly trains the image and text encoders to learn a multimodal feature space, maximizing the cosine similarity of N positive pairs while minimizing the distance among $N^2 - N$ negative pairs, achieving fine-grained image-to-text alignment.

Adaptive Dynamic Resolution Image Encoder: We use a ViT pretrained on 224×224 images [58]. Since MMSAR contains many high-resolution images, to prevent information loss and distortion during downsampling, we propose the AnyReader structure to flexibly handle arbitrary resolutions, with details provided in Section III-D. The image encoder utilizes either ViT or ResNet structures of different sizes. Experimental

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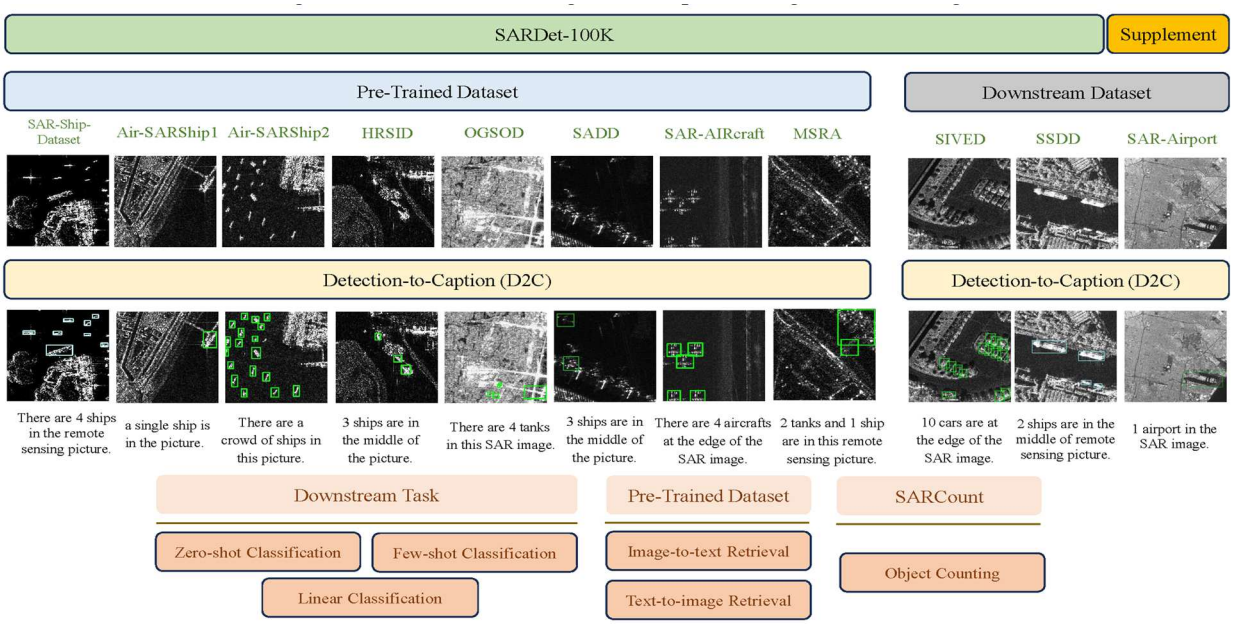


Fig. 2. Overview of the dataset of MMSAR. SARCLIP is trained on a diverse and comprehensive set of SAR image data, covering eight subsets from the SARDet-100K. For downstream tasks, we utilized SIVED, SSDD, and the supplementary SAR-Airport dataset. By applying the D2C algorithm, heterogeneous object detection annotations were converted into diverse captions, as illustrated above. To validate the generalization capability and robust performance of SARCLIP, we conduct extensive evaluations across six major downstream tasks, including the newly proposed object counting benchmark.

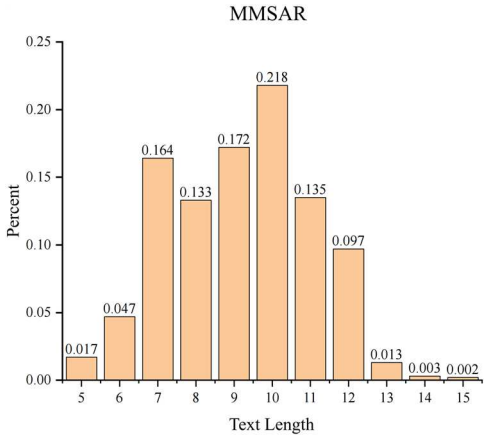


Fig. 3. Text length distribution of MMSAR.

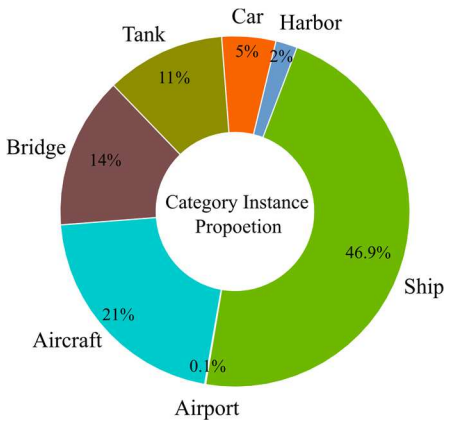


Fig. 4. Category instance distribution of MMSAR.

comparisons show that ViT-L-16 performs best on downstream tasks, so we chose ViT-L-16 as the foundation for SARCLIP, which defaults to using this architecture.

In this article, it produces a total of (224/16) $\times$ (224/16) $+$ 1 = 197 tokens, with a [CLS] token added at the beginning of the sequence to represent global image information.

Text Encoder: The text encoder is a Transformer architecture modified from the BERT model [5]. As a base model, we use a 120M, 12-layer, 768-width, and 12-head Transformer structure. Text is first tokenized to convert strings into discrete numerical ID spaces using a 49 152 size vocabulary generated by the byte pair encoding (BPE) algorithm [59]. The token embeddings are then used to map the discrete space into a continuous vector space, thereby enabling richer contextual interaction and more accurate semantic understanding during training. Each sentence includes a [CLS] token, and bidirectional attention is applied to comprehensively encode textual features. The final layer's vector at the [CLS] position effectively summarizes the semantic representation of sentences. A linear projection layer subsequently maps the textual features into a shared multimodal space, precisely aligned with corresponding image features.

Contrastive Learning: With the text and image features, we use InfoNCE loss [21] to align image-to-text pairs and maximize the diagonal positive sample similarity. $D = \{(x_i^T, x_i^I)\}_{i=1}^N$ represents N image-to-text pairs; and g^T and g^I denote the text encoder and image encoder, respectively. $g^T(x_i^T) = s_i^T \in \mathbb{R}^{1\times d}$ and $g^I(x_i^I) = s_i^I \in \mathbb{R}^{1\times d}$, in which s_i^T and s_i^I, respectively, represent the latent features of encoded text and image, with d as the feature dimension. As shown in (1), the training process simultaneously optimizes a bidirectional objective. τ represents a learnable temperature parameter that controls the

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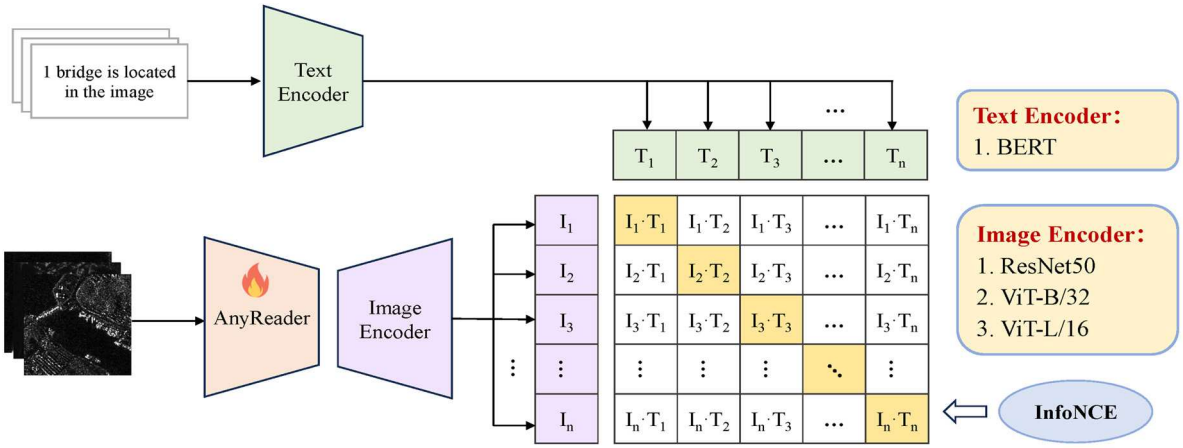


Fig. 5. Structure of SARCLIP. It mainly consists of three parts: a text encoder that extracts textual information, an adaptive dynamic resolution image encoder that can extract features from images of any size, and a text-to-image similarity matrix trained on correct text-to-image pairs using InfoNCE loss.

range of logits output by the softmax function. The first term corresponds to the text-to-image loss, while the second term represents the image-to-text loss

$$\mathscr{L}_{\text{InfoNCE}} = -\frac{1}{2N}\left(\sum_{i=1}^{N}\log\frac{\exp\left(s_i^T\cdot s_i^I/\tau\right)}{\sum_{j=1}^{N}\exp\left(s_i^T\cdot s_j^I/\tau\right)} + \sum_{i=1}^{N}\log\frac{\exp\left(s_i^I\cdot s_i^T/\tau\right)}{\sum_{j=1}^{N}\exp\left(s_i^I\cdot s_j^T/\tau\right)}\right). \quad (1)$$

InfoNCE loss provides two benefits as follows.

1) *Cross-Modal Understanding:* The model learns fine-grained correspondences between images and text, achieving a high similarity score $s_i^T \cdot s_i^I$, which corresponds to the (x_i^T, x_i^I) image-to-text pair while suppressing other negative samples. Generally, accurate alignment of representations can significantly enhance downstream performance in cross-modal retrieval tasks.
2) *Grouped Clustering:* Unimodal representations with similar semantic information tend to cluster more easily. Accurate representation grouping leads to strong single-modal recognition performance, such as linear classification.

D. AnyReader

The core idea of AnyReader is to use a low-resolution encoder to process high-resolution images. To avoid distortion and information loss caused by resizing, AnyReader is designed to handle images of arbitrary resolution. As shown in Fig. 6, it mainly consists of the adaptive cropping, abstractor and compression, and 2-D position embedding modules. The input image is first preprocessed using shape-adaptive cropping. The resulting subimages and the original image are then passed through the visual encoder and extractor. To capture the spatial relationships between subimages, 2-D position embedding is introduced.

Adaptive Cropping: First, we predefine grids of various shapes, with each grid corresponding to a subimage. $\{G = (m_h \times m_w)\,|\,m_h * m_w \le N, m_h \le 4, m_w \le 4\}$, where m_h is the number of rows and m_w is the number of columns in grid G. The maximum value of N is 16. To select the most suitable grid, we aim to ensure that the grid fits the original image resolution while also ensuring that each subgrid best fits the encoder's shape. Therefore, we propose an adaptive function, as shown in the following equation:

$$S_{\text{IOU}}(m_h, m_w) = \frac{\min\left(W_o, W_p * m_w\right)\times\min\left(H_o, H_p * m_h\right)}{W_o * H_o + m_w W_p * m_h H_p} \quad (2)$$

where (W_o, H_o) represents the given image size and (W_p, H_p) represents the image size of the pretrained. The fitness between the cropped subimage and the pretrained image is measured using the intersection over union, and the optimal grid size is determined by the following formula:

$$m_h^*, m_w^* = \text{argmax}\left(S_{\text{IOU}}(m_h, m_w)\right) \quad (3)$$

where m_h^* and m_w^* are the optimal segmentation ratios. Then, we split the image (W_o, H_o) into subimages, adjusting each subimage to the shape size (W_p, H_p) of pretrained encoders. To preserve the global structure of the image, the original image (W_o, H_o) is also resized to the specified size to feed into the encoder.

Abstractor and Compression: All subimages are processed in parallel through the ViT encoder, resulting in features $F \epsilon \mathbb{R}^{N\times(H^**W^*+1)\times d}$, where 1 represents the CLS token, $N = m_h^* \times m_w^* + 1$, and d represents the dimension of the visual feature. We extract the CLS tokens from each subimage to form the global image feature $F^* \epsilon \mathbb{R}^{N\times 1\times d}$, which is passed through the visual compression and abstraction layer. A learnable query vector $Q \epsilon \mathbb{R}^{1\times d_l}$ is used to perform cross attention, and the global image information is compressed and fused, where d_l is consistent with the dimension of the text space. Generally, the visual abstractor further refines visual information, producing more semantically rich visual representations.

2-D Position Embedding: Due to the lack of positional information between subimage blocks, the visual abstraction and fusion process may overlook the relative positional relationships between objects, making it difficult to align with text

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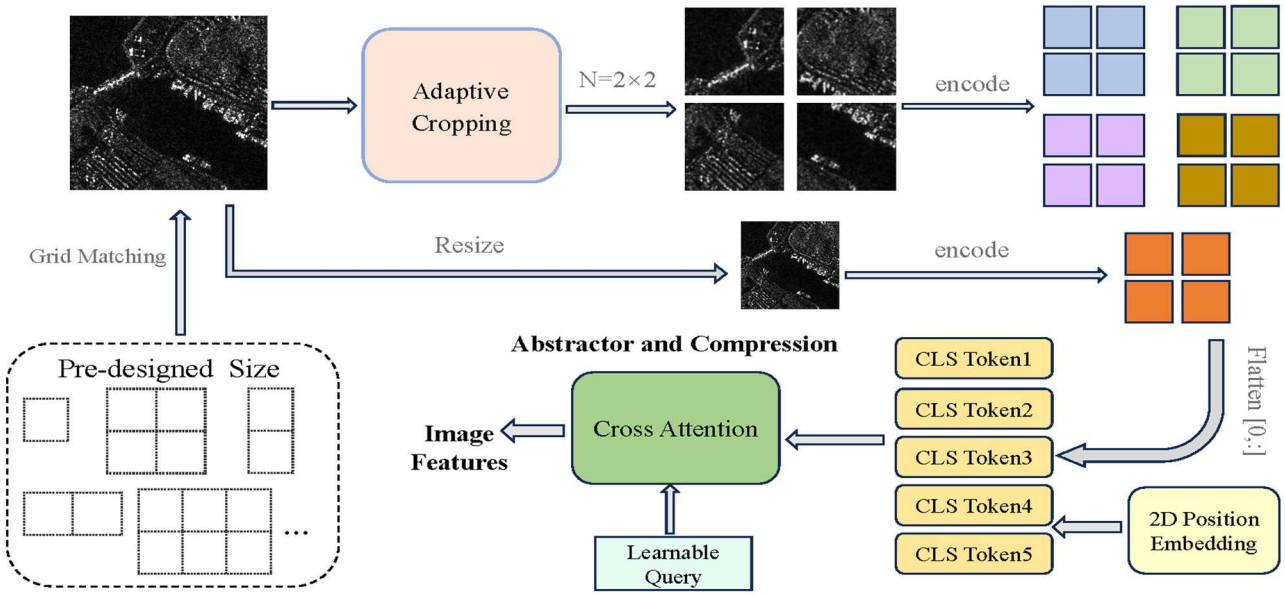


Fig. 6. Structure of AnyReader. The adaptive cropping module splits images of different resolutions into a specified size, such as $N = 2 \times 2$. The subimage blocks and the resized original image are then fed into the ViT-L-16 encoder separately. Finally, cross encoding is performed using a learnable query vector to extract image features that represent global information. The 2-D position embedding introduces the spatial relationship between different subimage blocks.

features [60]. Therefore, we introduce a 2-D position encoding module, assigning each subimage in the grid a positional index (i, j), where i represents the row coordinate and j represents the column coordinate. These indices are mapped to vector encodings through auxiliary embedding layers, as shown in the following equations:

$$\mathrm{PE}_{i,j}^{\mathrm{row}} = \mathrm{PositionEmbedding}_{\mathrm{row}}(i) \tag{4}$$

$$\mathrm{PE}_{i,j}^{\mathrm{column}} = \mathrm{PositionEmbedding}_{\mathrm{column}}(j) \tag{5}$$

$$\mathrm{PE}_{i,j} = \mathrm{PE}_{i,j}^{\mathrm{row}} + \mathrm{PE}_{i,j}^{\mathrm{column}} \tag{6}$$

where $\mathrm{PE}_{i,j} \epsilon \mathbb{R}^d$ represents the final positional encoding for each grid cell. The positional encoding $\mathrm{PE}_{i,j}$ is added to the global image feature $F^* \epsilon \mathbb{R}^{N \times 1 \times d}$ before performing cross attention. The generated spatially aware visual features will correspond to textual features at a finer granularity.

IV. Experiment

A. Implementation Details

1) Data: The final training dataset consists of 103 000 images, each accompanied by approximately three corresponding textual descriptions, resulting in a total of 296 750 image-to-text pairs. To prevent overfitting, we apply data augmentation techniques during training, including random ratio cropping to adjust image sizes, ultimately ensuring that each image matched the shape required by the pretrained encoder. To enhance dataset diversity and improve the model's robustness to images with different orientations, we apply random horizontal flipping and rotations at angles of 270°, 180°, 90°, and 0°. Gray padding is added to the excess parts of the images, and this data augmentation increases the model's rotational invariance.

2) Model: To determine the optimal model configuration for SARCLIP and validate the adaptability of our synthetic data across models of various scales and sizes, we employ three visual backbone architectures: ResNet50, a small-scale model with 38M parameters; ViT-B-32, a medium-scale model with 87M parameters; and ViT-L-16, a large-scale model with 304M parameters. The ResNet-50 used is the OpenAI version, where the max pooling is substituted with average pooling. In addition, an anti-aliasing rect-2 blur pooling is added to the uppermost layer, and the output layer replaces the original average pooling with an attention pooling. Attention pooling can be viewed as a single-layer multihead self-attention, with the input query derived from the image representation of global average pooling. ViT-B-32 comprises 12 attention heads and 12 layers, using a 32×32 window to divide the original image into fixed patches, each of which is transformed into a single token with a hidden dimension of 768. ViT-L-16 consists of 16 attention heads and 24 layers, utilizing a 16×16 window to partition the original image into patches, where the hidden dimension of each token is 1024. The text encoder is based on a Transformer architecture, with the maximum text length set to 77, similar to OpenAI CLIP. The InfoNCE loss is calculated by the [CLS] tokens from both the text encoder and the image encoder in parallel.

3) Continual Training and Settings: We train the SARCLIP model using mixed-precision FP16, which reduces memory usage while maintaining accuracy. Also, we use the pretrained weights from OpenAI to continue training on our own data. The optimizer used is Adam, and the learning rate strategy involves linear warm-up followed by cosine annealing. For ResNet50, ViT-B-32, and ViT-L-16, the learning rates are set to $4e^{-5}$, $4e^{-5}$, and $1e^{-4}$, and the batch sizes are set to 128, 128, and 32, respectively. We conduct distributed training using a single node with four 3090 Ti GPUs.

4) Benchmark: We evaluate the performance and generalization of the CLIP and SARCLIP models across six downstream

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TABLE III
PERFORMANCE OF CROSS-MODAL RETRIEVAL ON MMSAR (PRETRAINED MODEL)

Testing Dataset	Training Dataset	Training Samples	Method	Image Backbone		Text Backbone		Text to Image			Image to Text			MeanRecall
				Name	Params (M)	Name	Params (M)	R@1	R@5	R@10	R@1	R@5	R@10	
MMSAR (Pre-Trained)	MMSAR (Pre-Trained)	103,000	CLIP	ResNet50	38	Transformer	64	13.92	29.83	39.92	12.71	27.86	37.42	26.32
			CLIP	ViT-B-32	87	Transformer	64	17.84	38.71	48.83	15,57	37.14	47.43	32.57
			CLIP	ViT-L-16	304	Transformer	120	19.13	40.88	50.11	17.14	38.29	48.87	35.23
			SARCLIP	AnyReader (ViT-B-32)	103.5	Transformer	64	22.87	45.72	55.83	21.28	43.86	54.44	40.60
			SARCLIP	AnyReader (ViT-L-16)	343.5	Transformer	120	**23.83**	**46.78**	**58.21**	**21.68**	**45.87**	**56.58**	**42.3**

tasks, including image-to-text retrieval, text-to-image retrieval, zero-shot classification, few-shot classification, linear probing classification, and object counting.

B. Cross-Modal Image-to-Text Retrieval

We conduct experiments using different image and text encoder architectures based on the CLIP and SARCLIP models, with both training and testing datasets selected from MMSAR (pretrained). For cross-modal image-to-text retrieval during inference, we apply L2 normalization to the extracted image and text features. Retrieval is based on the dot product of feature matrices. For image-to-text, the image matrix is multiplied by the transpose of the text matrix, and vice versa for text-to-image. Recall is used as the evaluation metric, including R@1 (top-1), R@5 (top-5), R@10 (top-10), and mean recall. No specific data fine-tuning was applied.

Table III provides a detailed comparison of training sample sizes, backbone architectures, and parameter scales. For both image-to-text and text-to-image retrieval tasks, SARCLIP of all sizes consistently outperforms the CLIP models, thereby achieving state-of-the-art (SOTA) performance. Specifically, our proposed large-scale model, SARCLIP, comprises a 343.5M parameter AnyReader (ViT-L-16) and a 120M parameter text Transformer. Compared to the best-performing baseline, CLIP (ViT-L-16), it achieves a 7.07% higher mean recall. For text-to-image retrieval, it improves performance by 3.7% on R@1, 5.9% on R@5, and 8.1% on R@10. For image-to-text retrieval, it shows gains of 4.54% on R@1, 7.58% on R@5, and 7.71% on R@10. In addition, compared to the smaller SARCLIP model (103.5M parameter AnyReader of ViT-B-32 and a 64M text Transformer), it still achieves a 1.7% higher mean recall.

C. Counting Benchmark

The study [61] found that large-scale pretraining enables the CLIP model to excel in counting tasks ranging from 1 to 10. To evaluate whether SARCLIP possesses robust language understanding capabilities for target counting tasks on SAR images, we generate the SARCount benchmark dataset, specifically focusing on accurately counting 1–10 distinct targets. Fig. 7 visualizes several representative sample data, with text providing objective target counts, such as "a photograph of {six ships}." The dataset includes a total of 650 carefully constructed image-to-text pairs, sourced from MMSAR (pretrained), and it comprehensively covers five major categories: ships, tanks, bridges, ports, and aircraft. The figure presents both nonnumeric text representations and numeric text representations for enhanced diversity. The horizontal axis in the figure represents the prediction accuracy from top-1 to top-10 levels. We compare the largest parameter-scale models of SARCLIP and CLIP under the same settings. For each image, in addition to the text label of ground truth, the original number is replaced with nine other digits to automatically generate an additional nine alternative text descriptions. We then calculate image-to-text similarity scores and retrieve the highest probability text among the ten candidates. The number contained in the highest probability text is considered the final prediction result. Fig. 8 illustrates the prediction accuracy of both models from top-1 to top-10. It is evident that the overall curve of our SARCLIP consistently surpasses that of

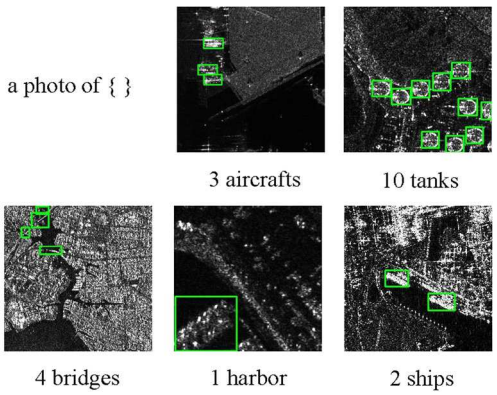


Fig. 7. New benchmark of SARCount. The figure above visualizes the captions and test images. The detected objects are labeled with green boxes.

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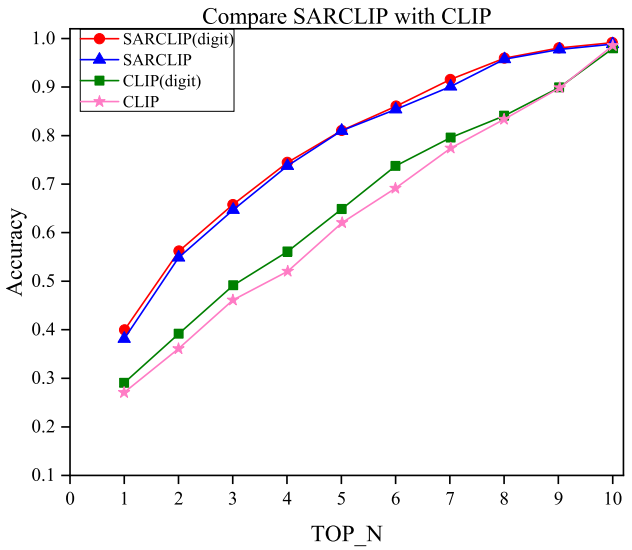


Fig. 8. Object detection and counting results of the CLIP and SARCLIP model on the SARCount benchmark.

TABLE IV
ZERO-SHOT CLASSIFICATION ACCURACY ON THREE DOWNSTREAM DATASETS

Method	Backbone	SAR-Airport	SSDD	SIVED	Average
CLIP	ResNet50	26.54	53.67	36.42	38.88
CLIP	ViT-B-32±Δ	32.16	64.75	46.18	47.69
SARCLIP	ViT-B-32±Δ	44.01 +11.85	73.24 +8.49	56.11 +9.93	57.78 +10.09
CLIP	ViT-L-16±Δ	38.69	69.84	52.33	53.62
SARCLIP	ViT-L-16±Δ	48.18 +9.49	75.51 +5.67	60.57 +8.24	61.42 +7.8

the CLIP model, with SARCLIP demonstrating significantly higher accuracy even at the top-6 level. In addition, to test the robustness to Roman and English numerals, "1"–"10" are replaced with "one"–"ten." The SARCLIP effectively adapts to this numerical variation, while the CLIP model showed fluctuations in counting accuracy.

D. Zero-Shot Classification

In this section, we conduct zero-shot classification experiments. The SAR-Airport, SSDD, and SIVED datasets are selected as downstream tasks, which are not included in the training set. In addition, we construct text label inputs using prompt templates, such as "a photograph of {class name}." Each image is compared with all text labels to calculate similarity, and the index with the highest probability corresponds to the predicted category.

Table IV shows the zero-shot classification results. Our SARCLIP model outperforms the CLIP baseline. With ViT-B-32 as the backbone, SARCLIP achieves an average classification accuracy 10.09% higher than CLIP, with approximately 9% improvement across all three datasets. Using the larger ViT-L-16 backbone, the average accuracy of SARCLIP is 7.8% higher than CLIP, with improvements of 9.49% on SAR-Airport, 5.67% on SSDD, and 8.24% on SIVED. In addition, when AnyReader is used as the backbone, the larger SARCLIP (ViT-L-16) outperforms SARCLIP (ViT-B-32), achieving an average accuracy of 61.42%, which is 3.64% higher.

TABLE V
FEW-SHOT ACCURACIES ON THREE DOWNSTREAM DATASETS OF CLASSIFICATION TASKS

Method	Backbone	SAR-Airport	SSDD	SIVED	Average
CLIP	ResNet50	74.40	83.45	76.88	78.24
CLIP	ViT-B-32	82.32	87.42	82.43	84.06
CLIP	ViT-L-16	85.65	89.12	84.61	86.46
SARCLIP	AnyReader (ViT-B-32)	88.46	**92.25**	87.84	88.95
SARCLIP	AnyReader (ViT-L-16)	**90.62**	91.54	**89.85**	**90.91**

E. Few-Shot Classification

In the field of few-shot learning, several studies have explored strategies to improve performance in SAR image classification and detection. For example, Chen et al. [62] leveraged feature attention and dynamic relation learning to model support-query interactions as a dynamic graph, thereby enabling task-adaptive learning under limited data.

Kirillov et al. [2] further addressed the cross-sensor scenario by proposing CS-FSDet, which incorporates the multiuncertainty balanced dynamic alignment (MUBDA) and the attention cross interaction with class awareness (ACICA) modules. These approaches demonstrate the importance of integrating dynamic perception mechanisms to maintain strong generalization capabilities when training data are scarce and distributions are complex.

Building on this line of research, we investigate the adaptability of SARCLIP under few-shot conditions. Although SARCLIP outperforms the CLIP baseline in zero-shot classification and achieves satisfactory average accuracy, its performance on certain datasets, such as SAR-Airport, remains suboptimal. To address this, we developed a targeted few-shot fine-tuning strategy. Specifically, we freeze the majority of model parameters and only train selected critical layers: the last attention pooling layer in ResNet50, the last Transformer layer in the ViT encoder of CLIP, and the final cross attention and projection layers in the AnyReader architecture. In addition, the image-to-text alignment module in all models is activated for training. This design keeps computational cost manageable while significantly enhancing the model's performance on specific downstream tasks.

Table V shows the classification performance of the fine-tuned models on three downstream tasks. It can be seen that the fine-tuned SARCLIP models significantly outperform the CLIP model. The smaller SARCLIP (ViT-B-32) and the larger SARCLIP (ViT-L-16) models outperform the largest CLIP model by an average accuracy of 2.49% and 4.45%, respectively. Especially, the larger SARCLIP (ViT-L-16)

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TABLE VI
LINEAR PROBING CLASSIFICATION EXPERIMENTS ON THREE DOWNSTREAM DATASETS

Method	Backbone	Linear Classifier			
		SAR-Airport	SSDD	SIVED	Average
CLIP	ResNet50	93.38	94.43	92.58	93.43
CLIP	ViT-B-32	93.29	96.02	93.88	94.40
CLIP	ViT-L-16	94.57	96.45	94.24	95.09
SARCLIP	AnyReader (ViT-B-32)	95.16	97.50	94.14	95.60
SARCLIP	AnyReader (ViT-L-16)	**96.71**	**98.25**	**95.55**	**96.84**

performs exceptionally well. Although it is slightly lower than SARCLIP (ViT-B-32) on the SSDD dataset, its classification accuracy on all three datasets reaches around 90%.

F. Linear Probing Classifier

Finally, we return to traditional linear classification tasks. We load the pretrained model weights and use only the image encoder to extract image features, which are then passed into a new linear classifier for training. The classifier is a logistic regression model with a learning rate set to 0.8, and a cosine annealing strategy is used to update the learning rate. The optimizer is SGD, the loss function is cross-entropy loss, the regularization coefficient C is set to 0.316, and the maximum number of iterations is set to 1000.

Table VI shows the performance of our newly trained linear classifier. It is evident that the average accuracy of the SARCLIP model is still higher than that of the CLIP model, indicating that SARCLIP with the AnyReader structure has stronger image representation ability when handling high-resolution images. The average accuracy of the large-sized SARCLIP (ViT-L-16) reaches 96.84%, which is 3.41% higher than CLIP (ResNet50) and 1.75% higher than CLIP (ViT-L-16).

V. CONCLUSION

This article introduces MMSAR, the first vision–language paired dataset in the SAR domain, comparable in scale to the COCO dataset. A D2C algorithm is proposed that converts heterogeneous formats of object detection bounding boxes into captions, ensuring the richness and diversity of the textual information. In addition, we introduce the SARCLIP, the vision–language multimodal model in the SAR domain, and design the AnyReader architecture to dynamically handle SAR images of any resolution. Through large-scale pretraining, SARCLIP learns fine-grained vision–language correspondences and achieves robust visual representations with rich semantic supervision. Aligned vision–language features enable better performance on downstream tasks. We conducted comprehensive evaluations on five tasks, especially in the newly introduced SARCount benchmark, image-to-text retrieval, and zero-shot tasks, where the SARCLIP significantly outperforms the SOTA CLIP baseline.

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